

HSC EXE Packaging Guide

Step-by-step guide to build and package the Windows installer EXE for the desktop GIS application.

IT Whitelist URLs (Before Setup)

Client IT must allow outbound access to the following domains before running dependency installation and packaging commands:

```
https://registry.npmjs.org
https://registry.yarnpkg.com
https://yarnpkg.com
https://classic.yarnpkg.com
https://github.com
https://api.github.com
https://objects.githubusercontent.com
https://github-releases.githubusercontent.com
https://release-assets.githubusercontent.com
https://electronjs.org
https://www.electronjs.org
https://github.com/electron/electron/releases
https://github.com/electron-userland/electron-builder-binaries/releases
```

Reason: Package installations use npm/yarn registries, and Electron/electron-builder binaries are downloaded from GitHub release/CDN endpoints.

Optional (only if Android build/sync steps are also performed):

```
https://maven.google.com/
https://repo1.maven.org/maven2/
https://services.gradle.org/
```

Build Environment Prerequisites

Node.js: 22.12.0

Yarn: 1.22.22

Other package versions: Refer to package.json

```
# Verify versions
node -v
yarn -v

# Install yarn globally if missing
npm install -g yarn
```

TLS/HTTPS Requirement: Secure HTTPS traffic must be allowed by client IT/network policy to avoid dependency download failures.

Root Naming Requirement: In this document, the project root is referred to as 'hsc-android'.

Project Structure Assumed

```
hsc-android/  
├─ package.json  
├─ electron/  
├─ src/  
└─ ... (rest of desktop project files)
```

Important: Client receives only this desktop project folder, renamed as 'hsc-android'.

Offline Tiles Location (Windows)

The desktop application reads tiles from the current Windows user's Documents\tiles folder.

Default:
C:\Users\<YourUser>\Documents\tiles

Using environment variable:
%USERPROFILE%\Documents\tiles

Minimum required structure:

```
tiles/  
├─ style.json  
├─ sprites/  
├─ fonts/  
└─ ... referenced tile data
```

Important: style.json must exist at the root of Documents\tiles, otherwise map style loading fails.

Packaging Steps

Step 1: Navigate to Project Folder

```
cd "hsc-android"  
# OR  
cd "D:\any-folder\hsc-android"
```

Step 2: Install Dependencies

```
yarn install
```

Step 3: Package EXE

```
yarn package
```

Client Flow: Run 'yarn install' first, followed by 'yarn package'. Packaging handles build steps internally.

Generated Output Files

Installer EXE: release\HSC GIS Desktop Setup 1.0.0.exe

Unpacked EXE: release\win-unpacked\HSC GIS Desktop.exe

Success Check: After packaging, installer EXE must exist in the release folder and the installed application should launch without missing asset errors.

Troubleshooting – NSIS Cache Fix

```
cd "hsc-android"
```

```
Get-Process node,electron,electron-builder,signtool -ErrorAction  
SilentlyContinue | Stop-Process -Force
```

```
Remove-Item "$env:LOCALAPPDATA\electron-builder\Cache\nsis" -Recurse -Force -  
ErrorAction SilentlyContinue
```

```
Remove-Item "$env:LOCALAPPDATA\electron-builder\Cache\nsis-resources" -Recurse  
-Force -ErrorAction SilentlyContinue
```

```
Remove-Item ".\release" -Recurse -Force -ErrorAction SilentlyContinue
```

```
New-Item -ItemType Directory -Force ".\cache\electron-builder" | Out-Null  
$env:ELECTRON_BUILDER_CACHE = (Resolve-Path ".\cache\electron-builder").Path
```

```
yarn package
```

.gitignore Entries (Client Side)

Ensure the following entries exist to avoid committing generated files:

```
# Electron build artifacts  
dist-electron/  
release/  
  
# Electron builder local cache  
cache/  
  
# JVM crash/replay logs  
android/hs_err_pid*.log  
android/replay_pid*.log
```